

Making Impact Visible in Funding of Open Source Software: A Study of the Funders' Perspective

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Funding needed, but limited

- OSS is critical digital infrastructure in modern software supply chains
- Demand-side value estimated at **\$8.8 trillion** (Hoffmann et al., 2024)
- Growing mix of corporate, public and platform funders (e.g., Alpha-Omega, Sovereign Tech Agency, GitHub Sponsors)
- Yet demand for funding greatly exceeds supply
- We need more funding, and better efficiency and impact of existing funding



Impact is hard to capture

- Ecosystem scale and project uniqueness make it hard to identify critical infrastructure and assess needs
- Downstream effects of funding decisions are hard to evaluate
- Prior work: single cases (e.g., scikit-learn, Sovereign Tech Agency, Alpha-Omega) and toolkits for public funding
- **Gap:** no broader study of how funders conceptualize, measure and report impact



Research question

- **RQ: How do OSS funders conceptualize, measure, and report funding impact, and what implications does this have for sustaining funding and OSS?**
- **Funding impact:** the extent to which funders achieve their objectives in providing support
- Focus on how funders select projects, define objectives, measure outcomes and justify continued funding



Research approach

- **25 semi-structured interviews** (30–60 min) across 19 organizations; targeted and snowball sampling until saturation
- Thematic analysis in three coding stages, with an audit trail
- **Workshop at FOSDEM 2026** (16 participants) for member checking and refinement
- Interview guide and workshop brief available on: <https://figshare.com/s/a8d07e164f0aff6b2344>



Who we talked to



25 interviews · 16 workshop participants · **23 organizations** across both

12

Funders

Alpha-Omega	Mozilla MOSS
Bloomberg FOSS Fund	Open Source Technology Improvement Fund
Chan Zuckerberg	Open Technology Fund
Google	Prototype Fund
Indeed	Sovereign Tech Agency
Microsoft	UNICEF Ventures

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Recipients

Internet Security Research Group
cURL
Drupal Association
Eclipse Foundation
Erlang Ecosystem Foundation
Probabl
Reproducible Builds
Rust Foundation

3

Research

Atlantic Council
CHAOSS community
Fraunhofer

Key insights

- Impact is mostly measured via short-term milestones and qualitative narratives
- It depends heavily on personal funder–recipient relationships
- Long-term sustainability, risk mitigation and human impacts are rarely measured
- Practices are shaped by implicit strategies, funding structures and reporting needs
- Result: ad-hoc, project-specific approaches



Six aspects of funding impact



1 Vision and strategy

What do funders seek to achieve, and why?

Often informal and person-dependent

2 Impact logic

How is funding expected to produce impact?

Theories of change are implicit and unshared

3 Operationalization

How is impact turned into objectives?

Shaped by funding instruments; deliverable-based

4 Types of impact

Which dimensions of change count?

Heterogeneous and hard to aggregate

5 Measurement approach

How is impact evidenced?

Ad hoc; favors what is easy to measure

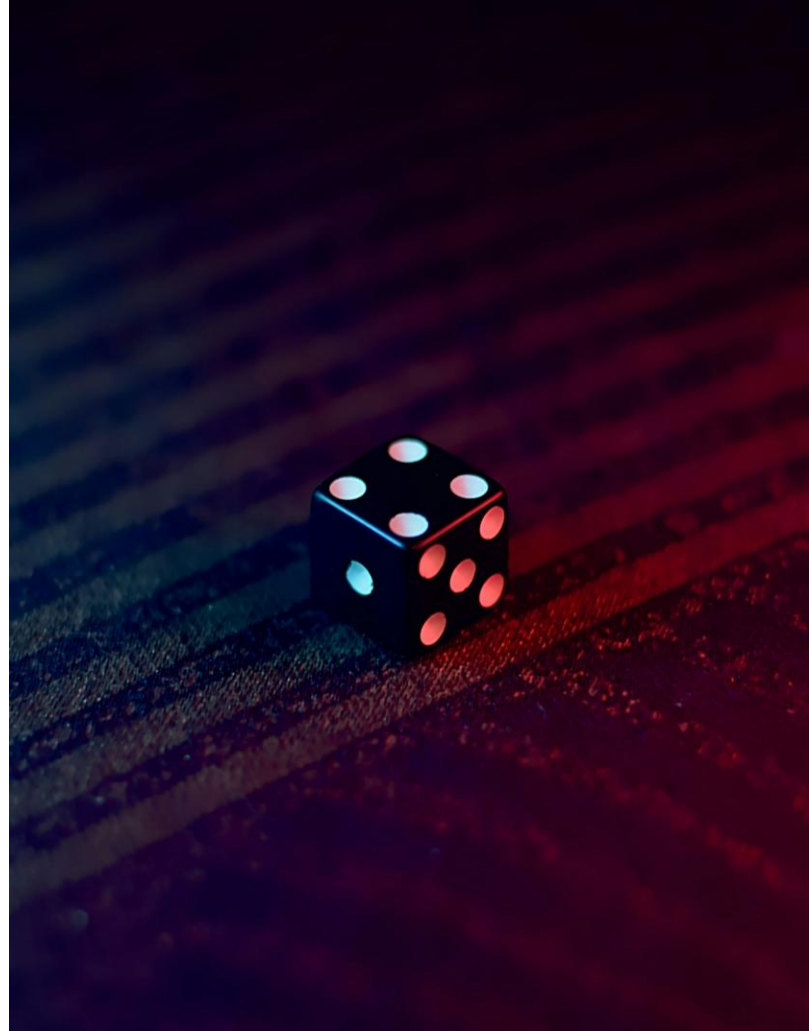
6 Impact reporting

How is impact made visible and credible?

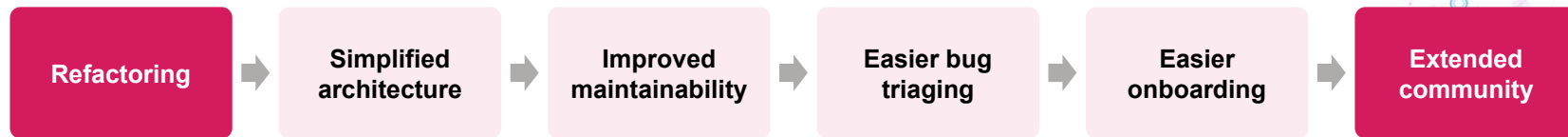
Translation and legitimization work

Vision and strategy

- Corporate funders: dependencies, internal OSS culture, influence in key technologies
- Foundations and governments: ecosystem security, resilience, digital infrastructure
- Some programs are partly charity, PR or employee engagement
- **Strategy is often tied to individuals, not institutionalized**
- When a champion or maintainer leaves, relationships and funding may reset



Implicit impact logic



Example chain from funded activity (short-term) to sustainability impact (long-term)

Proposed in the workshop: Goal–Signal–Metric

Goal

from the funder's strategy

Signal

what indicates progress

Metric

how to measure it

“We need to always be able to tell a story about, how are we contributing to the short-term goals and medium-term goals in our own internal theory of change”

— I5, funder (senior program manager)

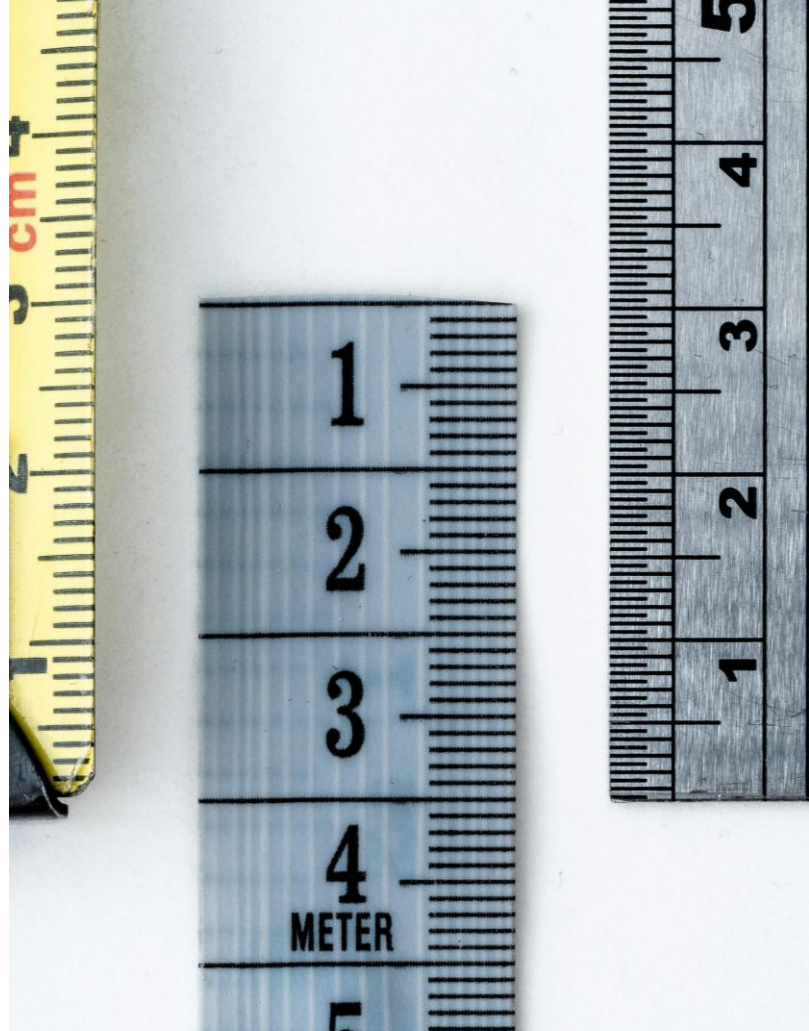
Money shapes measurement

- **Goal-focused:** milestones met, issues closed, audits addressed; easy to justify
- **Risk-based:** impact as risk avoided, but no vocabulary for things that didn't go wrong
- *"It's easy to say, 'Oh, we funded fixing this zero day,' but it's hard to say we prevented having more zero days." — I17*
- **Funding structures** (grants, contracts, stipends) scope what can be measured: deliverables, not ecosystem health



Types of impact

- **Financial ROI:** a latent requirement, asked for when funding is questioned
- **Security and sustainability:** vulnerabilities, audits, fuzzing; emerging SLA-style demand
- **Downstream use and adoption:** dependencies and reach; a multiplier for risk
- **Innovation and public good:** open release as ROI; Digital Public Goods Standard
- No coherent way to relate or aggregate these across portfolios



Measurement approach

- No shared framework; each project is “its own snowflake”
- Qualitative narratives preferred; quantitative metrics (e.g., CHAOSS) support the story
- Baselines: admission surveys, OpenSSF Scorecard, before/after security audits
- Infrastructure is measurable; people (burnout, succession) stay invisible until failure
- Long-term effects are bound by grant timelines and tracked informally



Reporting as translation

- Reported at fixed intervals, mostly at the end of funding
- Reports are read: *“Roughly 85% of our web traffic directly goes to the reports.” — I6*
- Narratives are case-specific; the “larger picture” is hard to show
- Audiences differ: corporate backers, ministries, internal sponsors
- So reports mix metrics, milestones, case studies and technical narratives



Discussion



1 **Bias toward short-term, visible work**

Binary, deliverable views miss effects that materialize later; risk-based scenarios give a systemic view

3 **Misaligned strategy, design and logic**

Goal–Signal–Metric (cf. GQM) is rarely explicit; MOSS a mature exception

5 **Reporting as translation**

A shared language to explain, e.g., what 20% more funding buys

2 **Human infrastructure: the missing unit**

Burnout, toxicity and governance matter as much as vulnerabilities fixed

4 **Meta-frameworks, not universal metrics**

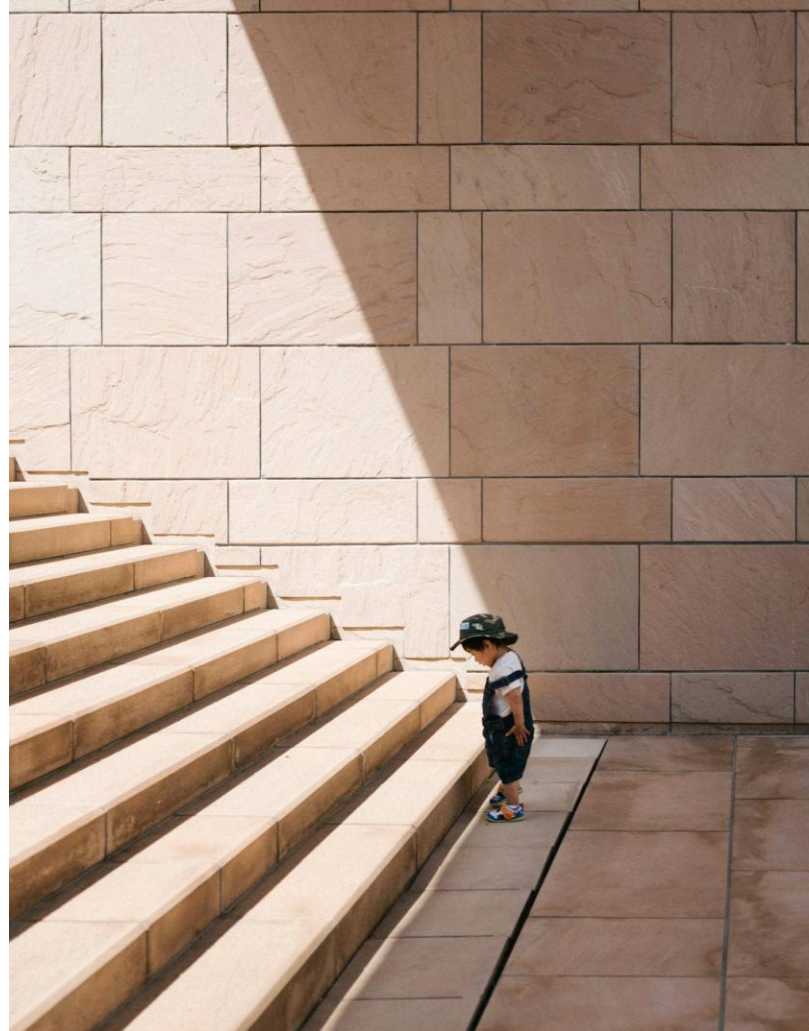
A shared interface for federated funding, respecting each funder's “dialect”

6 **From “built” to “relied upon”**

Supply-chain reliability and resilience framing links to SLAs and policy

Implications for funders

- **Make the impact logic explicit:** Goal → Signal → Metric, tied to strategy
- **Co-define goals** with recipients and set a baseline at admission
- **Track human infrastructure:** maintainer surveys, responsiveness, release cadence
- **Plan low-cost follow-up:** biannual calls, follow-up audits
- **Institutionalize** strategy and relationships to survive people turnover on both sides



Threats to validity



Credibility

Diverse roles and organization types; interviews triangulated with workshop; member checking

Transferability

Analytical, not statistical, generalization; mostly Euro-Atlantic funders

Dependability

Documented protocol, staged thematic analysis and audit trail; materials shared

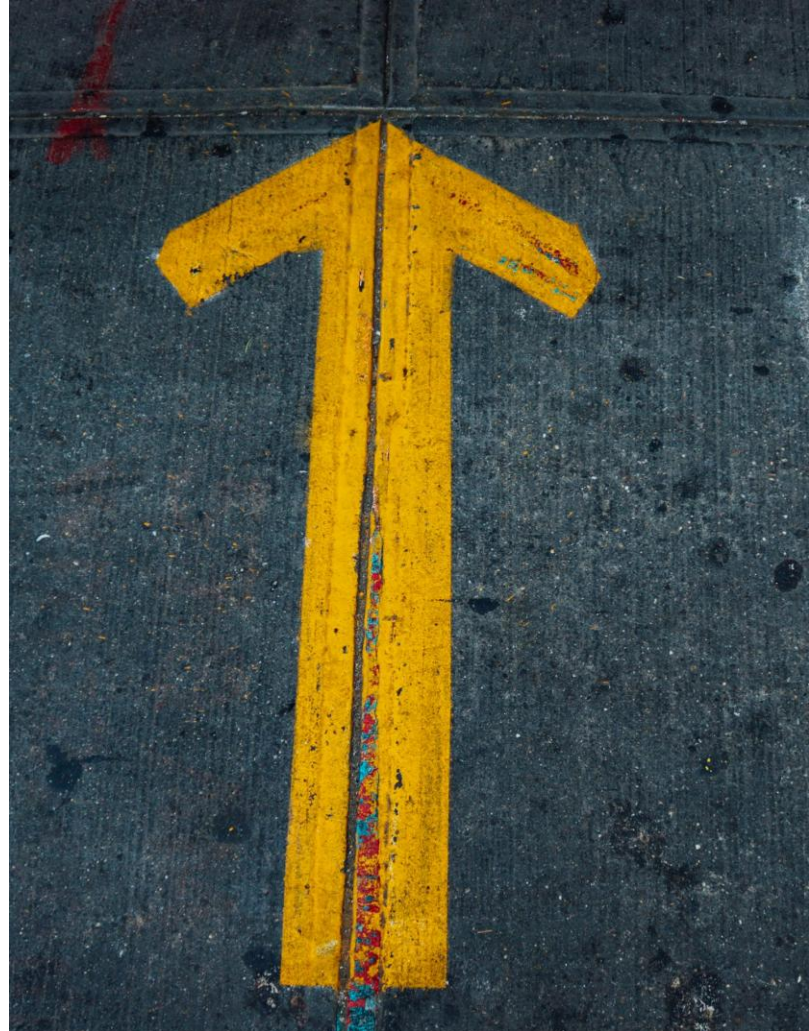
Confirmability

Authors embedded in OSS sustainability and CHAOSS; mitigated by grounding in quotes and member checking

Workshop organizations overlapped with interviewees, so it refined and contextualized findings rather than validating them independently (Guba, 1981).

Conclusions and future work

- Measuring and reporting funding impact is a shared, non-trivial challenge
- Gap between sustainability ambitions and how impact is operationalized
- Explicit impact models and shared meta-frameworks, not universal metrics
- Future work: a common language for a coordinated, federated funder ecosystem
- Priorities shift fast: security dominated interviews; AI now shapes funding



Thank you!



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Supplementary material: figshare.com/s/a8d07e164f0aff6b2344

